

ARCHITECTURE DOCUMENT

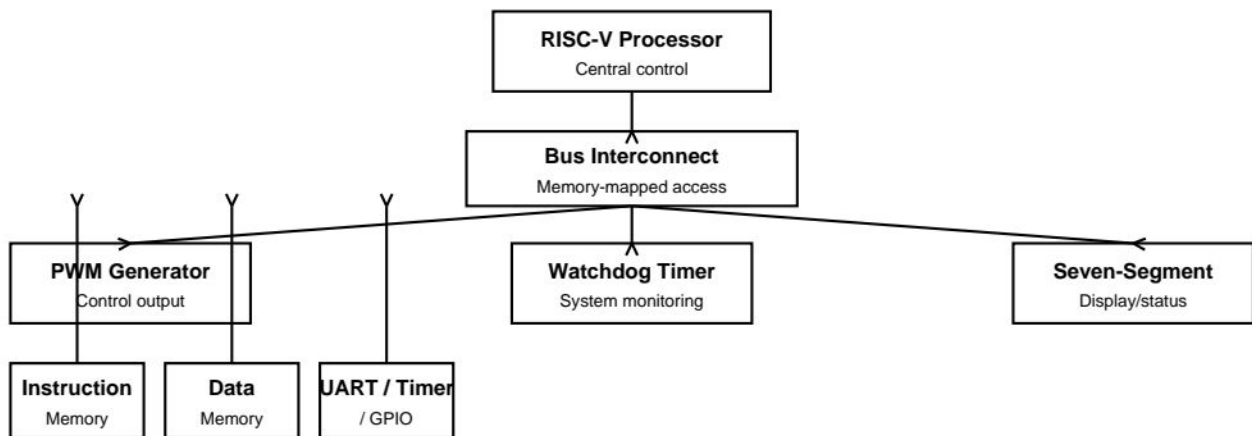
RISC-V Based SoC with PWM, Watchdog Timer and Seven-Segment Display Controllers

1. System Overview

The proposed SoC integrates a RISC-V processor with instruction memory, data memory, bus interconnect, UART, Timer and GPIO. Three additional IPs—PWM Generator, Watchdog Timer and Seven-Segment Display Controller—are integrated as processor-accessible peripherals. The RISC-V processor executes the application and controls the peripherals through the system interconnect.

Important: The signal names, widths, bus protocol and memory addresses in this document are preliminary. They must be changed to match the exact RISC-V and bus interface supplied by the faculty.

2. Proposed SoC Block Diagram



3. Major Blocks and Functions

Block	Function
RISC-V Processor	Executes the application and controls the SoC.
Instruction Memory	Stores program instructions.
Data Memory	Stores application data and intermediate values.
Bus Interconnect	Connects processor, memories and peripherals.
UART	Provides serial communication with a host.
Timer	Provides timing/counting functions.
GPIO	Provides general-purpose digital I/O.
PWM Generator	Generates a configurable PWM waveform.
Watchdog Timer	Monitors system activity and detects missed servicing.
Seven-Segment Controller	Drives segment signals for numerical/status display.

4. Proposed Memory Map

The following address map is illustrative and should be replaced with the final address allocation used by the supplied SoC.

Address Range	Block	Access	Purpose
0x0000_0000–0x0000_0FFF	Instruction Memory	R	Program storage
0x0000_1000–0x0000_1FFF	Data Memory	R/W	Data storage
0x0000_2000–0x0000_20FF	UART	R/W	Communication registers
0x0000_2100–0x0000_21FF	Timer	R/W	Timer registers
0x0000_2200–0x0000_22FF	GPIO	R/W	GPIO registers

Address Range	Block	Access	Purpose
0x0000_2300–0x0000_23FF	PWM	R/W	PWM control/status
0x0000_2400–0x0000_24FF	Watchdog	R/W	Watchdog control/status
0x0000_2500–0x0000_25FF	Seven-Segment	R/W	Display control/data

5. Interface Specifications

5.1 Common SoC / Memory-Mapped Interface

Signal	Dir.	Width	Purpose
clk	Input	1	System clock
rst_n	Input	1	Reset signal; polarity to match supplied design
addr	Master→Slave	32	Address
wdata	Master→Slave	32	Write data
rdata	Slave→Master	32	Read data
read_en	Master→Slave	1	Read request
write_en	Master→Slave	1	Write request
ready	Slave→Master	1	Transfer completion
resp	Slave→Master	2	Response/status, if required
irq	Slave→Master	1	Optional peripheral interrupt

5.2 PWM Generator

Signal	Dir.	Width	Purpose
pwm_enable	Input	1	Enable PWM
pwm_period	Input	16	PWM period/count value
pwm_duty	Input	16	Duty-cycle/count value
pwm_out	Output	1	Generated PWM waveform
bus_addr	Input	32	Peripheral address
bus_wdata	Input	32	Write data
bus_rdata	Output	32	Read data
bus_we / bus_re	Input	1/1	Write/read controls

5.3 Watchdog Timer

Signal	Dir.	Width	Purpose
wdt_enable	Input	1	Enable watchdog
wdt_reload	Input	16	Timeout/reload value
wdt_service	Input	1	Service/kick watchdog
wdt_timeout	Output	1	Timeout indication
wdt_reset_out	Output	1	Optional reset request
bus_addr	Input	32	Peripheral address
bus_wdata	Input	32	Write data
bus_rdata	Output	32	Read data

5.4 Seven-Segment Display Controller

Signal	Dir.	Width	Purpose
seg_enable	Input	1	Enable display
seg_data_in	Input	4	Digit value
seg_out	Output	8	Segment control signals
seg_digit_sel	Output	N	Digit select for multi-digit display
bus_addr	Input	32	Peripheral address
bus_wdata	Input	32	Display/control data

Signal	Dir.	Width	Purpose
bus_rdata	Output	32	Read-back/status data

6. Peripheral Register Map

PWM Generator

Offset	Register	Bits	Description
0x00	CONTROL	[0]	PWM enable
0x04	PERIOD	[15:0]	PWM period
0x08	DUTY	[15:0]	PWM duty value
0x0C	STATUS	[0]	PWM active/status

Watchdog Timer

Offset	Register	Bits	Description
0x00	CONTROL	[0]	Watchdog enable
0x04	TIMEOUT	[15:0]	Timeout value
0x08	SERVICE	[0]	Service/kick command
0x0C	STATUS	[1:0]	Active/timeout status

Seven-Segment Controller

Offset	Register	Bits	Description
0x00	CONTROL	[0]	Display enable
0x04	DATA	[31:0]	Display data
0x08	STATUS	[0]	Controller status

7. Bus Protocol Details

The proposed architecture uses memory-mapped peripheral access. The processor issues a read or write request with an address and, for writes, data. The bus interconnect decodes the address and routes the transaction to the selected memory or peripheral.

Write operation: Processor places address and data → write control is asserted → address decoder selects the target IP → IP updates its register → transfer completion is returned.

Read operation: Processor places address → read control is asserted → address decoder selects the target IP → IP returns register data → transfer completion is returned.

Protocol note: If the faculty-provided processor uses AXI4/AXI4-Lite, AHB-Lite, Wishbone or another specific protocol, the actual implementation must use that protocol's exact handshake and channel signals. The generic signals above are for the architecture document only.

8. System Operation / Application Flow

Step	Operation
1	RISC-V starts and initializes the peripherals.
2	Processor configures PWM period and duty cycle.
3	Processor enables and loads the Watchdog Timer.
4	Processor writes numerical/status information to the Seven-Segment Controller.
5	Processor periodically services the watchdog during normal operation.
6	PWM waveform, watchdog status and display behavior are verified.
7	A timeout test is performed by intentionally stopping watchdog servicing.

9. Verification Plan

Block	Tests	Expected Result
PWM	Enable/disable; 25%, 50%, 75% duty cycle; period change	Correct PWM waveform
Watchdog	Serviced normally; service stopped	No timeout / timeout respectively
Seven-Segment	Values 0–9; enable/disable	Correct segment pattern
Bus	Read/write peripheral registers	Correct address decode and data transfer
Complete SoC	Run RISC-V application	Processor controls all three IPs correctly

10. Expected Outcome

A functioning RTL-level RISC-V SoC in which the processor communicates with and controls the PWM Generator, Watchdog Timer and Seven-Segment Display Controller through the system interconnect. Individual IP testbenches and complete SoC simulation will be used to demonstrate correct operation.

Before implementation: obtain the faculty-provided RISC-V RTL, bus/interface specification, and any existing memory, UART, Timer and GPIO modules. Then replace the preliminary signal names, widths and address map in this document with the exact project values.